

Data and text mining

A non-negative matrix factorization based method for predicting disease-associated miRNAs in miRNA-disease bilayer network

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Abstract

Motivation: Identification of disease-associated miRNAs (disease miRNAs) is critical for understanding disease etiology and pathogenesis. Since miRNAs exert their functions by regulating the expression of their target mRNAs, several methods based on the target genes were proposed to predict disease miRNA candidates. They achieved only limited success as they all suffered from the high false-positive rate of target prediction results. Alternatively, other prediction methods were based on the observation that miRNAs with similar functions tend to be associated with similar diseases and vice versa. The methods exploited the information about miRNAs and diseases, including the functional similarities between miRNAs, the similarities between diseases, and the associations between miRNAs and diseases. However, how to integrate the multiple kinds of information completely and consider the biological characteristic of disease miRNAs is a challenging problem.

Results: We constructed a bilayer network to represent the complex relationships among miRNAs, among diseases and between miRNAs and diseases. We proposed a non-negative matrix factorization based method to rank, so as to predict, the disease miRNA candidates. The method integrated the miRNA functional similarity, the disease similarity and the miRNA-disease associations seamlessly, which exploited the complex relationships within the bilayer network and the consensus relationship between multiple kinds of information. Considering the correlation between the candidates related to various diseases, it predicted their respective candidates for all the diseases simultaneously. In addition, the sparseness characteristic of disease miRNAs was introduced to generate more reliable prediction model that excludes those noisy candidates. The results on 15 common diseases showed a superior performance of the new method for not only well-characterized diseases but also new ones. A detailed case study on breast neoplasms, colorectal neoplasms, lung neoplasms and 32 other diseases demonstrated the ability of the method for discovering potential disease miRNAs.

Availability and implementation: The web service for the new method and the list of predicted candidates for all the diseases are available at <http://www.bioinfolab.top>.

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Supplementary information: [Supplementary data](#) are available at *Bioinformatics* online.

1 Introduction

MicroRNAs (miRNAs) are small non-coding RNAs that regulate the expression of their mRNA targets through RNA cleavage or translational repression (Bartel, 2004; Chatterjee and Grosshans, 2009; He and Hannon, 2004). The dysregulation of miRNAs can cause developmental defects and contributes to progression of various diseases (Calin and Croce, 2006; Meola et al., 2009; Sayed and Abdellatif, 2011). Hence identifying disease-associated miRNAs (disease miRNAs) can provide novel insights into the genetic causes and consequences of complex diseases.

Aided by large scale deep sequencing, computational prediction of disease miRNAs can provide reliable miRNA candidates for further functional analysis in disease studies. Several methods have been developed for predicting disease miRNAs, which fall into two main categories. The methods in the first group extensively exploited the negative regulation of miRNAs on their target mRNAs (Bartel, 2004). They first identified the sets of target genes based on the complementarity between miRNA sequences and the sequences of putative target genes. The disease miRNA candidates were then inferred according to the similarities or the interactions between the target genes and the known disease-related genes (Jiang et al., 2010; Li et al., 2011; Shi et al., 2013). However, the experimentally validated target genes are so scanty that they fail to support the methods effectively. Hence the target prediction programs, such as TargetScan (Lewis et al., 2003) and PITA (Kertesz et al., 2007), have been adopted to obtain the majority of target genes. Due to the high false positive rate of target prediction results (Bartel, 2009; Liu et al., 2014; Ritchie et al., 2009), it is difficult for the methods of the first group to achieve excellent prediction accuracy.

In the second category, as functionally related miRNAs are usually involved in similar diseases (Bandyopadhyay, 2010; Goh et al., 2007; Lu et al., 2008), the functional similarity of two miRNAs was measured based on their associated diseases successfully (Wang et al., 2010). A functional similarity network of miRNAs was further constructed and denoted as *MiRNet*. Several methods were proposed to prioritize the miRNA candidates for a specific disease (Chen, 2012; Xuan et al., 2015) via random walks on *MiRNet*. Similarly, Chen's method inferred the disease candidates related to a specific miRNA via random walks on the disease network (Chen and Zhang, 2013). HDMP exploited the k most similar neighbors and the distribution of known disease miRNAs to infer the miRNA candidates (Xuan et al., 2013). These methods relied on a seed set of miRNAs that have already been related to the specific disease and therefore are not effective on the new diseases without any known related miRNAs. Recently, the information about diseases was introduced into the prediction methods to make them applicable to all the diseases, especially to the new ones (Chen and Yan, 2014; Liu et al., 2016; Xuan et al., 2015). However, Chen et al. established the separate objective functions for the miRNA network and the disease network respectively, which did not integrate the multiple kinds of information about miRNAs and diseases completely. Xuan et al. and Liu et al. concentrated on the prediction for the single disease and ignored the correlation between the candidates related to different diseases.

We propose and develop a novel prediction method based on non-negative matrix factorization and refer to it as *DMPred*. *DMPred* focuses on the following three important aspects. First, it is well known that miRNAs with similar functions tend to be associated with similar diseases and vice versa. As a result, the miRNA

functional similarity, the disease similarity and the miRNA-disease associations are consistent with each other. Considering the consensus relationship, *DMPred* integrates the multiple kinds of information about miRNAs and diseases seamlessly. Second, in terms of two similar diseases, their associated miRNA candidates are correlated rather than independent. Hence *DMPred* predicts their respective candidates for all the diseases simultaneously instead of for a single disease. Third, as only a small number of miRNAs are relevant to a specific disease (Kosik, 2006; Shi et al., 2013), the predicted associations between miRNAs and diseases should be sparse. *DMPred* takes the sparseness characteristic into account, which contributes to the generation of more reliable prediction model that excludes the noisy candidates.

2 Materials and methods

Our goal is to develop a global method that is able to simultaneously predict their respective associated miRNA candidates for all the diseases. We first constructed a bilayer network of miRNAs and diseases to represent the complex relationships between them. A novel prediction method based on non-negative matrix factorization with sparseness constraints was proposed specifically for the network.

Let S be the set that contains all the diseases. For a specific disease $d \in S$, the known d -related miRNAs are referred to as the *labeled nodes*, and the remaining miRNAs which have no information of relevance to d , are the *unlabeled nodes*. As the unlabeled nodes may potentially be associated with d , we correlate an unlabeled node u_i with an association score $S(u_i, d)$. The higher $S(u_i, d)$, the more u_i is likely to be associated with d . All the unlabeled nodes are ranked by their scores and the top ranked nodes are regarded as the promising d -related candidates.

2.1 Disease similarity measurement

The disease similarity quantifies how similar they are from the perspectives of disease semantics and symptom. The similarity of two diseases are calculated based on their common semantic annotations and shared disease symptoms.

Disease semantic similarity. We calculated the semantic similarity between diseases by using the existing measurement (Wang et al., 2010). Each disease is represented with a directed acyclic graph (DAG) which contains all the annotation terms related to the disease. Figure 1 shows the DAGs of two diseases 'Liver Neoplasms (LN)' and 'Stomach Neoplasms (SN)'. The DAG of a disease like LN is denoted as $DAG(LN) = (T_{LN}, E_{LN})$, where T_{LN} is a set that includes all the ancestor nodes of LN and LN node itself, and E_{LN} is a set of edges connecting these nodes. Each node t ($t \in T_{LN}$) has its semantic contribution, which is calculated by

$$D_{LN}(t) = \begin{cases} 1 & \text{if } t = LN \\ \max\{\Delta \cdot D_{LN}(t') | t' \in \text{children of } t\} & \text{otherwise} \end{cases}, \quad (1)$$

where Δ is an semantic contribution adjustment factor for the edges linking node t with its child t' . As suggested in the literature (Wang et al., 2010), it is set to 0.5. The overall semantic value of disease LN, $DV(LN)$, is defined as

$$DV(LN) = \sum_{t \in T_{LN}} D_{LN}(t). \quad (2)$$

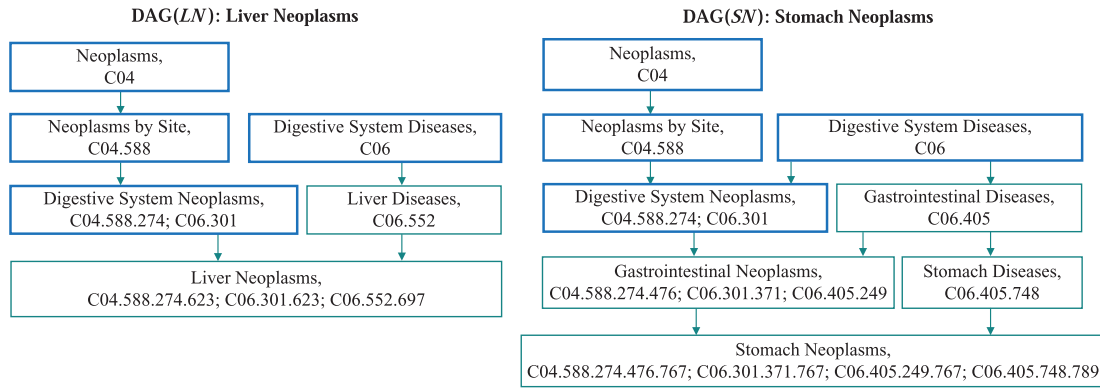


Fig. 1. The DAGs of the diseases *Liver Neoplasms* and *Stomach Neoplasms*. Each node contains one disease term and its identification numbers. The blue bold nodes are the common terms of these two diseases (Color version of this figure is available at *Bioinformatics* online.)

As two diseases sharing more terms in their DAGs are more similar, the semantic similarity between two diseases LN and SN is defined as

$$SS(LN, SN) = \frac{\sum_{t \in T_{LN} \cap T_{SN}} (D_{LN}(t) + D_{SN}(t))}{DV(LN) + DV(SN)}, \quad (3)$$

where $D_{LN}(t)$ and $D_{SN}(t)$ are the semantic values of term t related to diseases LN and SN , respectively. The semantic similarity of two diseases ranges between 0 and 1.

Disease phenotypic similarity. It is well studied that two diseases sharing more common phenotypes (signs and symptoms) are often more similar. *Hoehndorf et al. (2015)* measured the phenotypic similarities between diseases by integrating the phenotype data from the OMIM database (*Hamosh et al., 2005*) and the Orphanet database (*Weinreich et al., 2008*) and the phenotype ontology information. The phenotypic similarity of two diseases also ranges between 0 and 1.

Integrating disease semantic and phenotypic similarity. Let $PS(A, B)$ be the phenotypic similarity between two diseases A and B . In order to incorporate their semantic and phenotypic similarities, the similarity of A and B , $DS(A, B)$, is defined as follows,

$$DS(A, B) = \alpha SS(A, B) + (1 - \alpha) PS(A, B), \quad (4)$$

where $\alpha \in [0, 1]$ is a trade-off parameter determining the importance of the semantic similarity and it is set to 0.5 in our experiments. The values of the disease similarity range between 0 and 1.

2.2 MiRNA similarity measurement

The functional similarity of two miRNAs quantifies how similar their functions are. Based on the observation that miRNAs with similar functions are usually implicated in similar diseases, *Wang et al.* estimated the functional similarity of two miRNAs by measuring the similarity between their associated two groups of diseases (*Wang et al., 2010*). Consider *Figure 2a* as an example, miRNA m_a is associated with diseases d_1, d_2, d_3, d_4 and d_8 , and m_b is associated with d_1, d_2, d_5 and d_8 . The similarity between $DT_a = \{d_1, d_2, d_3, d_4, d_8\}$ and $DT_b = \{d_1, d_2, d_5, d_8\}$ is calculated as the functional similarity of m_a and m_b and denoted as $MS(m_a, m_b)$.

The miRNA similarity is calculated by using Wang's measurement method. We firstly compute the similarity between a disease, such as d_1 , and a group of diseases, such as DT_b . It is defined as follows,

$$S(d_1, DT_b) = \max_{1 \leq k \leq |DT_b|} (DS(d_1, d_k)), \quad (5)$$

where $d_k \in DT_b$. The similarity of two miRNAs, such as m_a and m_b , is obtained by calculating the similarity of DT_a and DT_b . It is defined as,

$$MS(m_a, m_b) = \frac{\sum_{1 \leq i \leq |DT_a|} S(d_i, DT_b) + \sum_{1 \leq j \leq |DT_b|} S(d_j, DT_a)}{|DT_a| + |DT_b|}, \quad (6)$$

where $|DT_a|$ and $|DT_b|$ is the numbers of diseases in DT_a and DT_b respectively. $S(d_i, DT_b)$ is the similarity between $d_i \in DT_a$ and the disease group DT_b , and $S(d_j, DT_a)$ is the similarity between $d_j \in DT_b$ and DT_a . The similarity of two miRNAs also ranges between 0 and 1.

2.3 Construction of miRNA-disease bilayer network

We construct a miRNA-disease bilayer network which contains two kinds of nodes (miRNAs and diseases) and three types of relationships (miRNA-miRNA similarity relationship and disease-disease similarity relationship, as well as miRNA-disease association relationship). The bilayer network is composed of a miRNA functional network, a disease network and the edges connecting the two networks. *Figure 2* demonstrates the workflow of constructing the network and its matrix representation.

Construction of MiRNet. The miRNA functional network (*MiRNet*) is constructed by connecting any two miRNAs whose functional similarity is more than 0. The topology structure of *MiRNet* and the functional similarities between miRNAs are captured by a weighted graph $G_M = (V_M, E_M, W_M)$. Each vertex $v_m \in V_M$ represents a miRNA, and an edge $e_m \in E_M$ connects two vertices indicating there is a functional link between them. The weight $w_m \in W_M$ of e_m quantifies the functional similarity degree of these two vertices. Let $M = [M_{ij}] \in \mathbb{R}^{N_m \times N_m}$ be an adjacency matrix of G_M . N_m is the number of miRNAs, $\mathbb{R}$ denotes the set of real numbers, and $\mathbb{R}^{N_m \times N_m}$ represents real coordinate space of $N_m \times N_m$ dimensions. M_{ij} is defined as follows,

$$M_{ij} = \begin{cases} MS(m_i, m_j) & \text{if there is an edge connecting } m_i \\ & \text{and } m_j \\ 0 & \text{otherwise} \end{cases}, \quad (7)$$

where $MS(m_i, m_j)$ is the functional similarity of m_i and m_j . Obviously, M is a symmetric matrix.

Construction of DisNet. In terms of the disease network (*DisNet*), its topology structure and the similarities between diseases are denoted by a weighted graph $G_D = (V_D, E_D, W_D)$ (*Fig. 2c*).

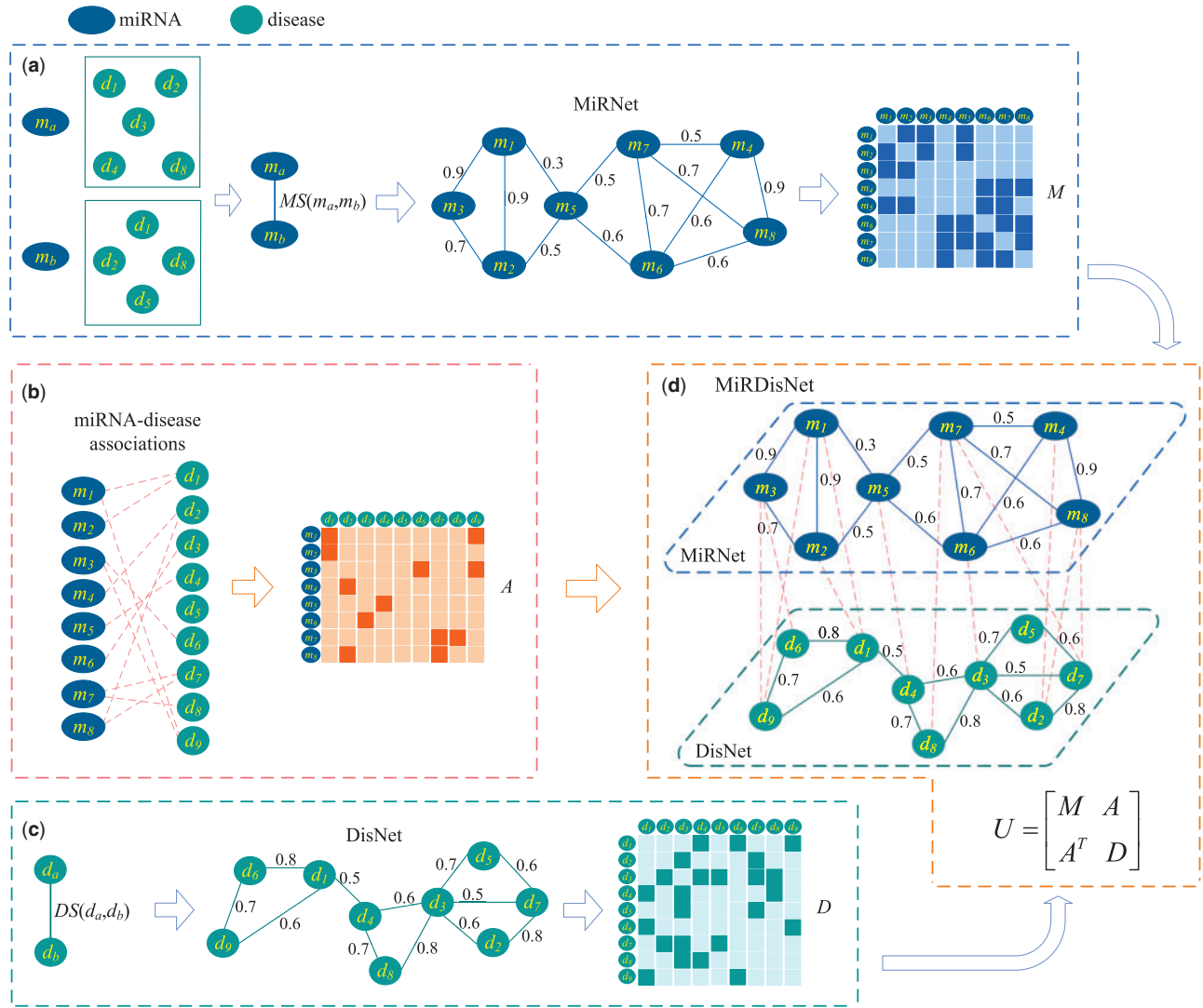


Fig. 2. Construction of a miRNA-disease bilayer network and its matrix representation. (a) Calculate the functional similarity of two miRNAs based on their associated diseases and construct a miRNA network (MiRNet) and its adjacent matrix M . (b) Construct an adjacent matrix A according to the known miRNA-disease associations. (c) Calculate the similarity of two diseases based on their common semantic annotations and shared symptoms, and construct a disease network (DisNet) and its adjacency matrix D . (d) Construct the miRNA-disease bilayer network (MirDisNet) and its adjacent matrix U

Each vertex $v_d \in V_D$ represents a disease, and an edge $e_d \in E_D$ captures the relation between two diseases. The weight w_d of edge e_d quantifies how similar two diseases are. Two disease nodes in DisNet are connected if their similarity is greater than 0. Let $D = [D_{ij}] \in \mathbb{R}^{N_d \times N_d}$ be an adjacency matrix of G_D . N_d is the number of diseases, and D_{ij} is defined as follows,

$$D_{ij} = \begin{cases} DS(d_i, d_j) & \text{if there is an edge connecting } d_i \\ & \text{and } d_j \\ 0 & \text{otherwise} \end{cases}, \quad (8)$$

where $DS(d_i, d_j)$ is the similarity of the diseases d_i and d_j . D is also a symmetric matrix.

Edges between MiRNet and DisNet. If a miRNA within MiRNet has been experimentally validated to be associated with a disease within DisNet, an edge is added to connect them (Fig. 2b). $A = [A_{ij}] \in \mathbb{R}^{N_m \times N_d}$ is a matrix representing the edges between MiRNet and DisNet. A_{ij} is 1 if miRNA m_i is associated with disease d_j , and it is 0 if their association has not been observed.

Finally, the miRNA-disease bilayer network, *MirDisNet*, is constructed by integrating MiRNet, DisNet and the edges between them (Fig. 2d). It is represented by a block adjacency matrix $U \in \mathbb{R}^{N \times N}$,

$$U = \begin{bmatrix} M & A \\ A^T & D \end{bmatrix}, \quad (9)$$

where $N = N_m + N_d$ and A^T is the transpose of A .

2.4 Non-negative matrix factorization based model for disease miRNA prediction

Let P_{ij} be the association score reflecting how likely miRNA m_i is associated with disease d_j . The score matrix for all the miRNAs and diseases is $P = [P_{ij}] \in \mathbb{R}^{N_m \times N_d}$, where each row corresponds to a miRNA (number of miRNAs is N_m) and each column corresponds to a disease (number of diseases is N_d). An objective function is established by integrating the multiple kinds of information within the bilayer network *MirDisNet*.

Modelling the edges between MiRNet and DisNet. As mentioned previously, A_{ij} is 1 if the association between miRNA m_i and disease d_j has been observed, and 0 if their association is unobserved. Furthermore, matrix A is very sparse and only a limited number of its entries are 1. In such cases, the model based on matrix factorization is usually optimized only over the actually observed entries (Natarajan and Dhillon, 2014).

Let Ω be the set of the observed miRNA-disease associations. In the case of $(i, j) \in \Omega$, the target association score between m_i and d_j , P_{ij} , needs to reflect as closely as possible to the observed association between the two, i.e. A_{ij} . $\|W \odot (P - A)\|_F^2$ is the deviation of the expected scores from the observations and can be computed as

$$\|W \odot (P - A)\|_F^2 = \sum_{i=1}^{N_m} \sum_{j=1}^{N_d} W_{ij} (P_{ij} - A_{ij})^2, \quad (10)$$

where $\|\cdot\|_F$ is the Frobenius norm of a matrix and $\odot$ is the Hadamard product. W is an observation indicator matrix where W_{ij} is set to 1 if $(i, j) \in \Omega$, and 0 otherwise.

Modelling MiRNet. For one thing, M_{ij} represents the actual functional similarity (FS) between m_i and m_j within MiRNet. For another, the well-known biological assumption means that if two miRNAs are associated with more similar diseases, they have higher FS. It indicates the miRNA FS is not only dependent on their associated diseases but also related to the similarities between diseases. Therefore, M can be factorized as PDP^T and $M \approx PDP^T$. The expected FS of m_i and m_j ($(PDP^T)_{ij}$) is calculated as,

$$(PDP^T)_{ij} = \sum_{r,s} P_{ir} P_{js} D_{rs}, \quad (11)$$

where P^T is the transpose of P . The i th row of P records the case that the diseases are associated with miRNA m_i and it is composed of P_{ir} ($1 \leq r \leq N_d$) which reflects the possibility that m_i is associated with the r th disease d_r . Similarly, P_{js} ($1 \leq s \leq N_d$) reflects the possibility that miRNA m_j is associated with disease d_s . D_{rs} is the similarity of d_r and d_s . As the expected FSs over all pairs of miRNAs should be as close as possible to their actual FSs, their squared loss is given as follows to quantify the difference between them,

$$\begin{aligned} \|(PDP^T) - M\|_F^2 &= \sum_{i,j} \left((PDP^T)_{ij} - M_{ij} \right)^2 \\ &= \sum_{i,j} \left(\sum_{r,s} P_{ir} P_{js} D_{rs} - M_{ij} \right)^2. \end{aligned} \quad (12)$$

Modelling DisNet. D_{ij} records the actual similarity between disease d_i and d_j within DisNet. On the other hand, the known biological premise also indicates that two diseases associated with functionally similar miRNAs is more similar. In other words, two groups of miRNAs associated with d_i and d_j and the FSs between these miRNAs are the latent factors affecting the similarity of d_i and d_j . Hence we factorize matrix D as $P^T M P$ and $D \approx P^T M P$. The expected similarity of d_i and d_j is $(P^T M P)_{ij}$, and its formal definition is as follows,

$$(P^T M P)_{ij} = \sum_{r,s} P_{ri} P_{sj} M_{rs}. \quad (13)$$

The i th column of P records the case that the miRNAs are associated with disease d_i and it is composed of P_{ri} ($1 \leq r \leq N_m$) which

reflects the possibility that d_i is associated with the r th miRNA m_r . In the same way, P_{sj} ($1 \leq s \leq N_m$) reflects the possibility that d_j is associated with the s th miRNA m_s . M_{rs} is the FS of m_r and m_s . The difference between the expected disease similarities like $(P^T M P)_{ij}$ and the actual similarities like D_{ij} can be calculated as follows,

$$\begin{aligned} \|(P^T M P) - D\|_F^2 &= \sum_{i,j} \left((P^T M P)_{ij} - D_{ij} \right)^2 \\ &= \sum_{i,j} \left(\sum_{r,s} P_{ri} P_{sj} M_{rs} - D_{ij} \right)^2. \end{aligned} \quad (14)$$

The Unified model. By integrating the information from the multiple components of MiRDisNet, we have the unified objective function as follows,

$$\begin{aligned} \min_{P \geq 0} L(P) &= \|W \odot (P - A)\|_F^2 + \lambda_m \|PDP^T - M\|_F^2 \\ &\quad + \lambda_D \|P^T M P - D\|_F^2, \end{aligned} \quad (15)$$

where λ_m and λ_D are regularization parameters adjusting the contribution of the latter two terms. As the association score between a miRNA and a disease should be greater than or equal to 0, the non-negative property of matrix P is enforced as a constraint.

In addition, since only a small number of miRNAs are associated with a specific disease, it seems reasonable that P should have a limited number of non-zero entries. So we add a sparse penalty term of P to the objective function as in (15),

$$\begin{aligned} \min_{P \geq 0} L(P) &= \|W \odot (P - A)\|_F^2 + \lambda_m \|PDP^T - M\|_F^2 \\ &\quad + \lambda_D \|P^T M P - D\|_F^2 + \theta \|P\|_1, \end{aligned} \quad (16)$$

θ is the regularization parameter and $\|\cdot\|_1$ denotes the ℓ_1 norm.

2.5 Optimization

As the function in (16) is not convex, it is impractical to get its globally optimal solution. We give an iterative algorithm based on coordinate descent to obtain its locally optimal solution. On the basis of the properties of the trace and Frobenius norm of matrix, the function $L(P)$ can be rewritten as,

$$\begin{aligned} L(P) &= \|W \odot (P - A)\|_F^2 + \lambda_m \|PDP^T - M\|_F^2 + \lambda_D \|P^T M P - D\|_F^2 \\ &\quad + \theta \|P\|_1 \\ &= \text{Tr}(W \odot (PP^T - PA^T - AP^T + AA^T)) \\ &\quad + \lambda_m \text{Tr}(PDP^T PD^T P^T - PDP^T M^T - MPD^T P^T + MM^T) \\ &\quad + \lambda_D \text{Tr}(P^T MPP^T M^T P - P^T MPD^T - DP^T M^T P + DD^T) + \theta \|P\|_1, \end{aligned} \quad (17)$$

where $\text{Tr}()$ denotes the trace of a matrix. Considering the nonnegative constraint on P , we introduce a Lagrange multiplier $\Psi = [\psi_{ij}] \in \mathbb{R}^{N_m \times N_d}$ and establish the Lagrange function as follows,

$$\begin{aligned} L(P, \Psi) &= \text{Tr}(W \odot (PP^T - PA^T - AP^T + AA^T)) \\ &\quad + \lambda_m \text{Tr}(PDP^T PD^T P^T - PDP^T M^T - MPD^T P^T + MM^T) \\ &\quad + \lambda_D \text{Tr}(P^T MPP^T M^T P - P^T MPD^T - DP^T M^T P + DD^T) \\ &\quad + \theta \|P\|_1 + \text{Tr}(\Psi P^T). \end{aligned} \quad (18)$$

After taking derivative of $L(P, \Psi)$ with respect to P and setting it to zero, we obtain

$$\Psi = 2W \odot (P - A) + 4\lambda_M(PDP^T PD - MPD) + 4\lambda_D(MPP^T MP - MPD) + \theta B, \quad (19)$$

where $B = [B_{ij}] \in \mathbb{R}^{N_m \times N_d}$ is a matrix whose elements are all 1. Both sides of Equation (19) are multiplied by P_{ij} and then the Karush-Kuhn-Tucker conditions ($\psi_{ij}P_{ij} = 0$) is used to get the following equation,

$$2W_{ij}(P - A)_{ij}P_{ij} + 4\lambda_M(PDP^T PD - MPD)_{ij}P_{ij} + 4\lambda_D(MPP^T MP - MPD)_{ij}P_{ij} + \theta B_{ij}P_{ij} = \psi_{ij}P_{ij} = 0. \quad (20)$$

Finally, according to the gradient decent algorithm (Tan and Févotte, 2009), the value of P_{ij} is updated by multiplying its current value with the ratio of the negative terms to positive terms of (20),

$$P_{ij}^{new} \leftarrow P_{ij} \frac{2W_{ij}A_{ij} + 4\lambda_M(MPD)_{ij} + 4\lambda_D(MPD)_{ij}}{2W_{ij}P_{ij} + 4\lambda_M(PDP^T PD)_{ij} + 4\lambda_D(MPP^T MP)_{ij} + \theta B_{ij}}. \quad (21)$$

Given an initial value of P , the solution of P can be obtained iteratively by using the above updating rule. The iterative process terminates when the difference between the values of $L(P)$ at the k th iteration and at the $(k+1)$ th is less than 10^{-7} . At last, in terms of disease d_j , P 's j th column contains the association scores between d_j and all the miRNAs including the labeled nodes and the unlabeled nodes. Since a higher score reveals a more possible association between a miRNA candidate and d_j , all the unlabeled nodes are ranked by their scores. The iterative algorithm of predicting disease miRNAs is demonstrated in Figure 3.

3 Results and discussion

3.1 Data preparation

The human miRNA-disease database (HMDD) has collected thousands of the miRNA-disease association pairs that had been confirmed by the biological experiments (Li et al., 2014a). As done in previous work (Chen and Chen, 2014; Wang et al., 2010; Xuan et al., 2015), we merged the redundant miRNA-disease associations that produce the same mature miRNAs and obtained 5090 distinct associations between 490 miRNAs and 326 diseases. The disease terms within the disease directed acyclic graphs and their hierarchies

Input: the miRNA-disease bilayer network and its adjacent matrix U ; the regularization parameters $\lambda_M \geq 0$, $\lambda_D \geq 0$, and $\theta \geq 0$

Output: the ranked miRNA candidates and their association scores with respect to each specific disease

- 1 Initialize matrix $P^{(0)}$ randomly with the values in the range $[0, 1]$
- 2 While not converged
- 3 update $P^{(k+1)}$ by using the multiplicative rule:

$$P_{ij}^{new} \leftarrow P_{ij} \frac{2W_{ij}A_{ij} + 4\lambda_M(MPD)_{ij} + 4\lambda_D(MPD)_{ij}}{2W_{ij}P_{ij} + 4\lambda_M(PDP^T PD)_{ij} + 4\lambda_D(MPP^T MP)_{ij} + \theta B_{ij}}$$

4 End While

5 For the j th column of the final P , corresponding to the disease d_j ($1 \leq j \leq N_d$)

6 All the unlabeled nodes are ranked by their scores

7 The unlabeled nodes with higher rankings are the more potential d_j -related miRNA candidates

8 End For

Fig. 3. Iterative algorithm for predicting the miRNA candidates associated with diseases

were acquired from the U.S. National Library of Medicine (MeSH, <http://www.ncbi.nlm.nih.gov/mesh>).

3.2 Performance evaluation metrics

To evaluate our approach and the state-of-the-art disease-miRNA-prediction methods, 5-fold cross validation was performed for the well-characterized diseases firstly. For a specific disease d that has known related miRNAs, the d -related miRNAs (labeled nodes) were randomly divided into 5 subsets, 4 of which were used for training a prediction model, while the left out subset was added into a dataset for testing. The testing dataset also contains all the miRNAs that have not been observed to be associated with d (unlabeled nodes). The labeled and unlabeled nodes are regarded as the positive samples and the negative ones, respectively. After the association scores of the testing samples are estimated, the samples will be ranked by their scores. The higher the positive samples are ranked, the better the prediction performance is.

We are also interested in evaluating the ability to correctly identify the miRNAs associated with the new diseases. As there is no any miRNA observed to be related to a new disease so far, the evaluation is simulated by using a disease with known related miRNAs, such as d , and removing all the d -related associations in the training process. In this way, d -related miRNA candidates are predicted by only exploiting the information about the remaining diseases. All the removed d -related miRNAs are taken as positive samples for testing.

Given a threshold δ , if the score of a labeled node is greater than δ , it is deemed as a correctly identified positive sample. If the score of an unlabeled node is smaller than δ , it is a successfully identified negative sample. To obtain a receiver operating characteristic (ROC) curve, the true positive rates (TPRs) and the false positive rates (FPRs) at various δ values are calculated,

$$TPR = \frac{TP}{TP + FN}, FPR = \frac{FP}{TN + FP}, \quad (22)$$

where TP and TN are the numbers of correctly identified positive and negative samples. FP and FN are the numbers of misidentified positive and negative samples. The area under the ROC curve (AUC) is used to measure the global performance of a prediction method.

In addition, the top section of prediction result is usually selected by the biologists to further validate with the wet-lab experiments, and the more accurate top k candidates contribute to the success of discovering the novel disease miRNAs. Therefore, the recall rates within top 30, 60, ... and 240 candidates are demonstrated, which reveals how many positive samples are successfully recovered within top k . Moreover, the proportion of misidentified negative samples and the proportion of total positive and negative samples identified correctly within top k ranking list are shown as well.

In the current miRNA-disease association data, most of diseases are only associated with several miRNAs, which results in lack of sufficient associations to evaluate the prediction performance. Hence we performed the cross-validation and simulation experiments on 15 well-characterized diseases each of which has at least 80 related miRNAs.

3.3 Comparison with other methods

As most of previous methods can only be applied to the well-characterized diseases, we estimated the prediction performances on these diseases and the new ones, respectively. Firstly, our method, DMPred, was compared with RWRMDA (Chen, 2012), Chen's

Table 1. Prediction results of DMPred and other methods for the well-characterized diseases

Disease name	AUC DMPred	RWRMDA	Chen's method	RLSMDA	MIDP	Liu's method
Acute myeloid leukemia	0.896	0.839	0.716	0.853	0.913	0.878
Breast neoplasms	0.940	0.785	0.653	0.832	0.838	0.847
Colorectal neoplasms	0.839	0.793	0.662	0.831	0.845	0.850
Glioblastoma	0.906	0.680	0.607	0.714	0.786	0.841
Heart failure	0.986	0.722	0.761	0.738	0.821	0.815
Hepatocellular carcinoma	0.902	0.749	0.613	0.794	0.807	0.835
Lung neoplasms	0.945	0.827	0.606	0.855	0.876	0.912
Melanoma	0.911	0.784	0.642	0.807	0.837	0.852
Ovarian neoplasms	0.928	0.882	0.644	0.909	0.923	0.898
Pancreatic neoplasms	0.915	0.871	0.684	0.887	0.945	0.899
Prostatic neoplasms	0.950	0.823	0.629	0.841	0.882	0.857
Renal cell carcinoma	0.899	0.815	0.627	0.839	0.862	0.820
Squamous cell carcinoma	0.901	0.819	0.676	0.849	0.870	0.877
Stomach neoplasms	0.901	0.779	0.628	0.797	0.821	0.827
Urinary bladder neoplasms	0.924	0.821	0.632	0.845	0.897	0.863

Table 2. A pairwise comparison with a paired *t*-test on the prediction results based on AUCs

	RWRMDA	Chen's method	RLSMDA	MIDP	Liu's method
<i>P</i> -value between DMPred and another method	2.9441e-06	1.8670e-12	4.1391e-05	8.4149e-04	7.3586e-05

method (Chen and Zhang, 2013), RLSMDA (Chen and Chen, 2014), MIDP (Xuan *et al.*, 2015) and Liu's method (Liu *et al.*, 2016) which are state-of-the-art prediction methods for the well-characterized diseases.

The regularization parameters of each method should be tuned to obtain its best performance. The parameters λ_M , λ_D and θ of DMPred were chosen from 1/10, 1/30, ..., 1/90, 1, 10, 30, ..., 90. The parameter r for both RWRMDA and Chen's method varied from 0.1 to 0.9. As its literature suggested, the parameters η_M and η_D of RLSMDA were set as 1 according to the prior knowledge, and w ranged from 0.1 to 0.9. The parameters r_Q and r_U of MIDP and the parameters γ , λ , δ and η of Liu's method ranged from 0.1 to 0.9. The performances of these methods obtained by using the optimum parameters are demonstrated in Table 1 ($\lambda_M = 1/70$, $\lambda_D = 1/10$ and $\theta = 1/20$ for DMPred, $r = 0.9$ for RWRMDA, $r = 0.8$ for Chen's method, $\eta_M = 1$, $\eta_D = 1$ and $w = 0.9$ for RLSMDA, $r_Q = 0.4$ and $r_U = 0.1$ for MIDP, $\gamma = 0.5$, $\lambda = 0.8$, $\delta = 0.9$ and $\eta = 0.1$ for Liu's method).

As shown in Table 1, the average AUCs of DMPred, RWRMDA, Chen's method, RLSMDA, MIDP and Liu's method on the 15 tested diseases are 91.63, 79.93, 65.20, 82.61, 86.15 and 85.81%, respectively. DMPred performed the best for most of these diseases and its average AUC is 11.7, 26.43, 9.02, 5.48 and 5.82% higher than the other methods, respectively. Note that MIDP and RWRMDA exploited the information of miRNA network, and Chen's method utilized the information of disease network. DMPred, RLSMDA and Liu's method not only concentrated on miRNA network but also on disease network, and they all achieved relatively better performances. Chen's method worked much worse than the other methods, primarily because it did not use the information of miRNA network. Therefore, the use and integration of information of miRNAs and diseases are essential. The improvement of DMPred over the existing methods is mainly due to its seamless integration of multiple kinds of information.

In addition, a paired *t*-test was performed to measure whether DMPred's AUCs across 15 diseases are significantly higher than

another method. The *p*-values are listed in Table 2. The statistical result indicates DMPred achieves significantly better performance than all of other methods at the significance level 0.05.

The higher recall value within top *k* ranking list means the more positive testing samples (real disease-related miRNAs) are identified successfully. The average recall values across 15 tested diseases within the top *k* candidates are shown in Figure 4. DMPred consistently outperformed the other methods at various *k* cutoffs, and ranked 49.6% of positive samples in top 30, 87.1% in top 90 and 97.8% in top 150. MIDP had the second-best accuracy and ranked 43.6% in top 30, 78.4% in top 90 and 90.9% in top 150. Liu's method ranked 41.8% in top 30, 77.2% in top 90 and 89.0% in top 150, which is worse than MIDP but better than RLSMDA (32.8, 73.4 and 86.9%). RWRMDA achieved inferior performance and its corresponding recall rates are 26.7, 68.4 and 83.7%. Chen's method ranked 8.5% in top 30, 33.7% in top 90 and 61.3% in top 150, which is still much worse than other methods.

Specificity measures the proportion of correctly identified negative samples accounting for all the negative samples. Thus, 1-specificity reflects the proportion of misidentified negative samples. A lower 1-specificity value on the top *k* ranking list means less negative samples are misidentified. As shown in Figure 5, DMPred yields the lowest 1-specificity values at different *k* cutoffs. In addition, accuracy is the proportion of true positive and negative samples that are identified correctly. DMPred also achieves higher accuracies than the other methods at various *k* values (Fig. 6).

To evaluate DMPred's performance for the new diseases without known related miRNAs, we performed the simulation experiments on the same 15 diseases as the ones in cross validation procedure. Unlike the cross validation experiments, all the associations related to a tested disease *d* were removed during training period. This operation ensured that predicting *d*-related candidates only utilized the association information of the remaining diseases and the miRNA and disease similarity information of MiRDisNet. All the removed *d*-related miRNAs were taken as the positive testing samples.

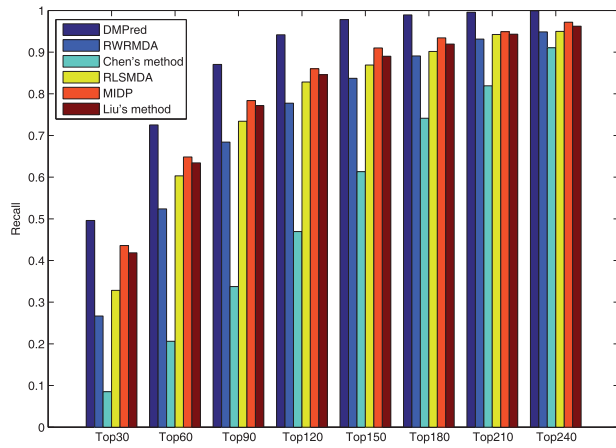


Fig. 4. The average recalls across all the tested diseases at different top k values

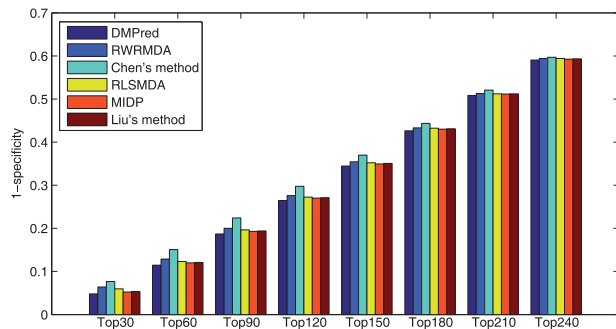


Fig. 5. The average 1-specificity values across all the tested diseases at different top k values

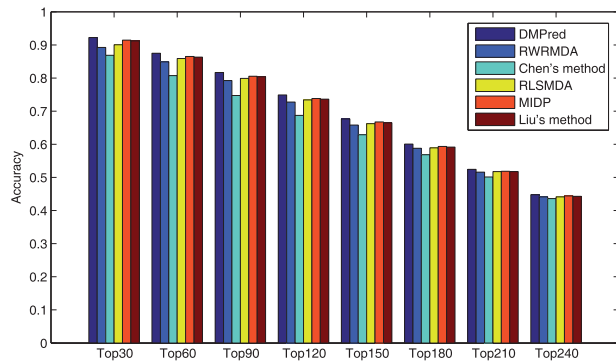


Fig. 6. The average accuracies across all the tested diseases at different top k values

As only RLSMDA, MIDP and Liu's method can be applied to the new diseases, DMPred was compared with them and the results are shown in Table 3. The average AUCs of DMPred, RLSMDA, MIDP and Liu's method across 15 diseases are 0.876, 0.837, 0.839 and 0.832. DMPred consistently performed the best for nearly all of 15 diseases, and MIDP achieved slightly better performance than RLSMDA for most of the diseases. The performance of Liu's method is relatively inferior. The primary reason is that DMPred considered the correlation between the candidates of various diseases, relative to MIDP and Liu's method. Meanwhile, DMPred integrated the multiple kinds of information within MiRDisNet

Table 3. Prediction results of DMPred and the other methods for the diseases whose respective associations were removed

Disease name	AUC			
	DMPred	RLSMDA	MIDP	Liu's method
Acute myeloid leukemia	0.868	0.852	0.860	0.864
Breast neoplasms	0.877	0.803	0.821	0.806
Colorectal neoplasms	0.856	0.812	0.829	0.815
Glioblastoma	0.868	0.831	0.833	0.828
Heart failure	0.895	0.792	0.814	0.801
Hepatocellular carcinoma	0.865	0.789	0.804	0.785
Lung neoplasms	0.921	0.897	0.874	0.878
Melanoma	0.876	0.817	0.825	0.817
Ovarian neoplasms	0.906	0.894	0.876	0.875
Pancreatic neoplasms	0.891	0.895	0.891	0.891
Prostatic neoplasms	0.854	0.844	0.829	0.827
Renal cell carcinoma	0.878	0.809	0.824	0.810
Squamous cell carcinoma	0.886	0.883	0.864	0.865
Stomach neoplasms	0.844	0.781	0.806	0.788
Urinary bladder neoplasms	0.860	0.852	0.836	0.832

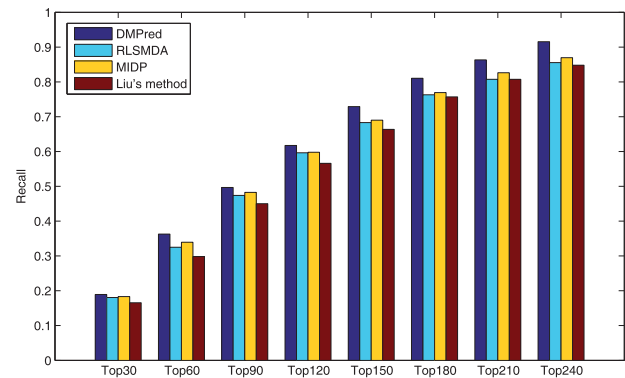


Fig. 7. The average recalls across all the simulated new diseases at different top k values

completely, relative to RLSMDA. In addition, the p -values of DMPred versus other three methods obtained by performing the paired t -test are $1.7635e-04$, $4.4946e-06$ and $9.4690e-06$. It confirms DMPred's performance is also significantly higher in terms of the new diseases. In addition, DMPred consistently had the highest average recall rates (Fig. 7), the lowest 1-specificity values (Fig. 8) and the highest accuracies (Fig. 9) on 15 tested diseases at different top k cutoffs.

3.4 Comparison with the prediction instance without sparse penalty

In order to validate the effect of exploiting sparseness characteristic, we further compared the prediction instance with sparse penalty (DMPred) and the one without penalty (DMPred*). The penalty item $\theta||P||_1$ was eliminated from the original objective function $L(P)$ to form a new function $L^*(P)$. The prediction instance based on $L^*(P)$ is referred to as DMPred*. As shown in Table 4, the left part demonstrates the results by performing 5-fold cross validation over 15 diseases, and the right part lists the results of simulation experiments after removing the associations related to the tested disease. DMPred achieves consistently higher prediction performances than DMPred* for the well-characterized diseases and the simulated new ones. In terms of the well-characterized diseases, AUC of DMPred increased at least by 1.1%, increased at most by 9.7% and increased

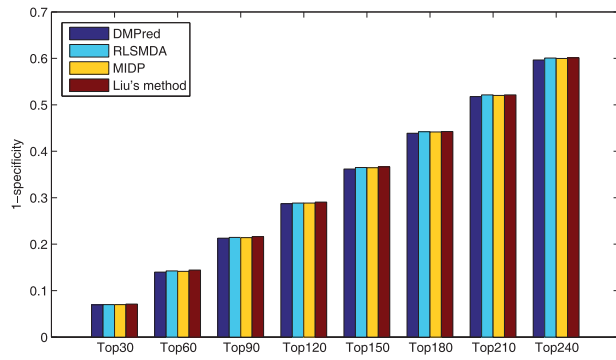


Fig. 8. The average 1-specificity values across all the simulated new diseases at different top k values

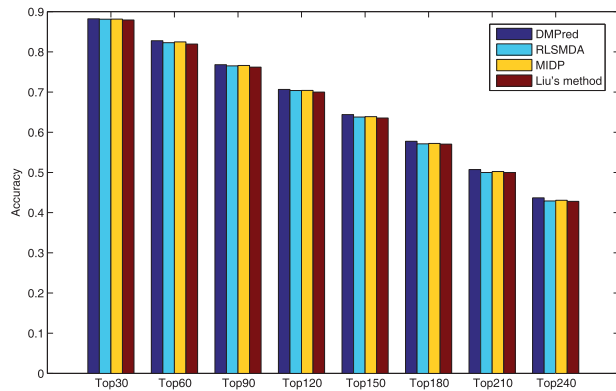


Fig. 9. The average accuracies across all the simulated new diseases at different top k values

by 5.2% on average. For the new diseases, AUC of DMPred increased at least by 1.4%, increased at most by 5.5% and increased by 4% on average. It indicates that introducing sparse penalty is effective for the improvement of the discriminative ability of prediction model.

3.5 Case studies: breast neoplasms, colorectal neoplasms, lung neoplasms and 32 diseases

To further demonstrate DMPred's ability to discover potential disease miRNA candidates for the well-characterized diseases, we executed the case studies on breast neoplasms, colorectal neoplasms and lung neoplasms. The top 50 candidates related to breast neoplasms are taken as examples and analyzed in detail.

First, a database named PhenomiR was established to demonstrate the miRNAs which have different expression in disease tissues relative to the normal ones (Ruepp *et al.*, 2010). PhenomiR contained 675 miRNAs that had been identified by analyzing the results of microarray experiments, northern blot experiments and PCR experiments. Similarly, the dbDEMC database included 607 miRNAs that had abnormal expression in 14 kinds of cancers through analysis of microarray datasets (Yang *et al.*, 2010). As shown in Table 5, 39 of 50 candidates are contained by PhenomiR and 39 candidates are included by dbDEMC, which indicates they have been upregulated or downregulated in breast cancer (malignant breast neoplasm).

Second, a miRNA-cancer association database, miRCancer, included the experimentally validated 878 associations between 236 miRNAs and 79 cancers (Xie *et al.*, 2013). The associations were

Table 4. Prediction results obtained by performing the prediction instance with sparse penalty (DMPred) and the one without sparse penalty (DMPred*)

Disease name	AUC			
	DMPred	DMPred*	DMPred	DMPred*
Acute myeloid leukemia	0.896	0.835	0.868	0.821
Breast neoplasms	0.940	0.874	0.877	0.850
Colorectal neoplasms	0.839	0.813	0.856	0.804
Glioblastoma	0.906	0.822	0.868	0.840
Heart failure	0.986	0.929	0.895	0.877
Hepatocellular carcinoma	0.902	0.805	0.865	0.842
Lung neoplasms	0.945	0.913	0.921	0.876
Melanoma	0.911	0.824	0.876	0.862
Ovarian neoplasms	0.928	0.910	0.906	0.851
Pancreatic neoplasms	0.915	0.904	0.891	0.808
Prostatic neoplasms	0.950	0.891	0.854	0.814
Renal cell carcinoma	0.899	0.855	0.878	0.858
Squamous cell carcinoma	0.901	0.824	0.886	0.837
Stomach neoplasms	0.901	0.830	0.844	0.801
Urinary bladder neoplasms	0.924	0.872	0.860	0.807

extracted from the published literatures by using text mining technique and then the dysregulation cases of the miRNAs were confirmed manually. miR2disease is also a database which contained manually curated miRNAs that had dysregulation in various diseases (Jiang *et al.*, 2009). Seven candidates are contained by miRCancer and 2 candidates are recorded by miR2disease. Therefore, they have been verified to be breast cancer-related miRNAs.

Finally, 9 candidates labeled with 'literature' are supported by the published literatures and the detailed interpretation is listed in Supplementary Table ST1. Several studies verified that 7 of 9 miRNAs were dysregulated in breast tumors versus normal breast tissues. In addition, hsa-mir-449b is a direct transcriptional target of E2F1 which is an important transcription factor relative to breast cancer (Yang *et al.*, 2009). Estrogen receptor-alpha (ER- α) has become one of the most important target in breast cancer therapeutics. Hsa-mir-302e was inferred to inhibit the expression levels of ER- α and negatively regulate ER- α -mediated signaling pathways in breast cancer (Li *et al.*, 2014b). Hence hsa-mir-449b and hsa-mir-302e are the promising breast cancer-related candidates.

In terms of colorectal neoplasms, the top 50 candidates are demonstrated in Supplementary Table ST2. Seven candidates were contained by PhenomiR and 40 candidates were included by dbDEMC to have abnormal expression in colorectal cancer. MiR2Disease and miRCancer respectively confirmed that the expression levels of 3 candidates and 13 candidates varied significantly between the colorectal tumors and normal colorectal tissues. Five candidates were supported by the literatures to be dysregulated in colorectal neoplasms.

The top 50 lung neoplasms-related candidates are listed in Supplementary Table ST3. PhenomiR and dbDEMC respectively identified 38 candidates and 43 candidates whose abnormal expressions have been found in lung cancer. miRCancer confirmed 25 candidates to have differential expression in lung tumors versus normal lung tissues. Ten candidates were verified by miR2disease to have been associated with the disease. A candidate was supported by the literature to be dysregulated in lung neoplasms.

In addition, since RLSMDA and MIDP concentrated on 32 new diseases to show their ability to effectively determine the potential candidates, DMPred was also applied to the diseases. The top 3 potential candidates for each disease were validated by the literatures

Table 5. The top 50 breast neoplasms-related candidates

Rank	MiRNA name	Description	Rank	MiRNA name	Description
1	hsa-mir-130a	PhenomiR, dbDEMC, miRCancer	26	hsa-mir-211	PhenomiR, dbDEMC, literature ¹
2	hsa-mir-99a	PhenomiR, dbDEMC, miRCancer	27	hsa-mir-212	PhenomiR, dbDEMC
3	hsa-mir-138	PhenomiR, dbDEMC, literature ¹	28	hsa-mir-449b	literature ²
4	hsa-mir-192	PhenomiR, dbDEMC	29	hsa-mir-181c	PhenomiR, dbDEMC
5	hsa-mir-142	PhenomiR, literature ¹	30	hsa-mir-144	PhenomiR, dbDEMC
6	hsa-mir-378a	PhenomiR, dbDEMC	31	hsa-mir-302f	literature ¹
7	hsa-mir-106a	PhenomiR, dbDEMC	32	hsa-mir-574	dbDEMC
8	hsa-mir-130b	PhenomiR, dbDEMC	33	hsa-mir-491	PhenomiR, dbDEMC
9	hsa-mir-15b	PhenomiR, dbDEMC	34	hsa-mir-184	PhenomiR, dbDEMC
10	hsa-mir-186	PhenomiR, dbDEMC	35	hsa-mir-208a	PhenomiR, dbDEMC
11	hsa-mir-542	literature ¹	36	hsa-mir-371a	PhenomiR
12	hsa-mir-150	PhenomiR, dbDEMC, miRCancer	37	hsa-mir-363	dbDEMC
13	hsa-mir-92b	literature ¹	38	hsa-mir-190a	literature ¹
14	hsa-mir-95	PhenomiR, dbDEMC	39	hsa-mir-376a	PhenomiR, dbDEMC
15	hsa-mir-330	PhenomiR, dbDEMC	40	hsa-mir-517a	PhenomiR, dbDEMC
16	hsa-mir-185	PhenomiR, dbDEMC	41	hsa-mir-503	dbDEMC
17	hsa-mir-99b	PhenomiR, dbDEMC, miRCancer	42	hsa-mir-302e	literature ²
18	hsa-mir-449a	PhenomiR, dbDEMC	43	hsa-mir-483	PhenomiR, dbDEMC
19	hsa-mir-98	PhenomiR, dbDEMC, miR2disease, miRCancer	44	hsa-mir-455	PhenomiR
20	hsa-mir-32	PhenomiR, dbDEMC	45	hsa-mir-650	dbDEMC
21	hsa-mir-370	PhenomiR, dbDEMC	46	hsa-mir-381	PhenomiR, dbDEMC
22	hsa-mir-181d	PhenomiR, dbDEMC, miR2disease	47	hsa-mir-30e	PhenomiR
23	hsa-mir-494	PhenomiR, dbDEMC, miRCancer	48	hsa-mir-744	dbDEMC
24	hsa-mir-196b	PhenomiR, dbDEMC, miRCancer	49	hsa-mir-134	PhenomiR, dbDEMC
25	hsa-mir-372	PhenomiR, dbDEMC	50	hsa-mir-518b	PhenomiR

Note: (1) ‘PhenomiR’ means that a miRNA was identified to have abnormal expression in breast cancer by analyzing the results of microarray experiments, northern blot experiments and PCR experiments. (2) With analysis of the microarray datasets, a miRNA is considered to have different expression levels in breast cancer compared with normal tissues. This kind of miRNAs is labeled with ‘dbDEMC’. (3) ‘miR2Disease’ means that a miRNA is contained by the manually curated miRNA-disease association database, miR2Disease and confirmed to be associated with breast cancer. (4) ‘miRCancer’ means an association between a miRNA and breast cancer is included by the database miRCancer. (5) ‘literature¹’ means that there is a published literature to support that a miRNA is upregulated or downregulated in breast neoplasm, compared with normal breast tissues. (6) ‘literature²’ means that a miRNA is related to some important factors affecting the development of breast neoplasms.

and the related database. Forty-six miRNA-disease associations were supported by the literatures and one association has been recorded by miR2Disease (see details in [Supplementary Table ST4](#)). On the whole, the case studies indicate that DMPred has powerful ability to discover potential candidates for not only well-characterized diseases but also new ones.

3.6 Predicting novel disease-related miRNAs

After having confirmed its prediction performance by cross validation and simulation experiments, as well as case studies, we further applied DMPred to all the diseases including the ones with known related miRNAs and the new ones. All the known miRNA-disease associations were taken as training data to predict the novel disease-related miRNAs. The potential candidates for all the diseases are listed in [Supplementary Table ST5](#).

4 Conclusions

A novel method based on non-negative matrix factorization with sparseness constraints, DMPred, was developed for predicting disease miRNAs. DMPred integrated multiple kinds of information within the miRNA-disease bilayer network seamless, which exploited the consensus relationship between them completely. Furthermore, DMPred took the correlation between the candidates of various diseases into account and predicted their respective candidates for all the diseases at the same time. In addition, incorporating the sparseness characteristic of miRNA-disease associations also

contributed to the improvement of prediction performance. The results of cross validation and simulation experiments on 15 common diseases confirmed the superiority of DMPred for the well-characterized diseases and the new ones. The case studies on 3 well-characterized diseases and 32 new diseases further demonstrated DMPred’s ability to discover the potential candidates. DMPred will be useful in screening the promising candidates for subsequent studies concerning their involvement in the etiology and pathogenesis of diseases.

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